Simplicity Refinements and Value Elicitation: What Outperforms the BDM on Initial Play?

Alexander L. Brown* Jinliang Liu† Michael Tsoi‡

March 25, 2026

Abstract

Experimental economics research has consistently documented systematic deviations from weakly dominant strategies. Economic theory has introduced behavioral refinements informed by these findings. This paper tests three such refinements in a standardized environment: initial play with value elicitation mechanisms. We utilize a unique design that juxtaposes each refinement with a closely related, "unrefined," baseline elicitation mechanism. Specifically, using a Becker-DeGroot-Marschak (BDM) mechanism, a dynamic descending-price list, and a descending-price clock mechanism, we test behavioral refinements that simplify the payoff function (contingency-by-contingency), the game structure (game-structure obviousness), and the dominant strategy (obvious dominance), respectively. While no refinement significantly outperforms its baseline, the price list format—regardless of whether it satisfies game-structure obviousness—substantially outperforms both the BDM and clock mechanisms in terms of the rate of subjects' dominant strategy play.

Keywords: dominant strategy, elicitation mechanisms, experiments, game-structure obviousness, obvious strategy-proofness.

JEL Codes: C72, C91, D82.

*Department of Economics, Texas A&M University, 4228 TAMU, College Station, TX 77843, United States of America. Email: alexbrown@tamu.edu.

†Corresponding author. School of Economics, Shandong University, 27 Shanda South Road, Jinan, Shandong 250100, China. Email: jinliang.liu.econ@outlook.com.

‡United States Postal Service, 475 L'Enfant Plaza SW, Washington, DC 20260, United States of America. Email: tsoi2019@gmail.com.

1 Introduction

For over sixty years, economics has championed the use of dominant strategy mechanisms for the elicitation of private information. Because these strategy-proof mechanisms contain equilibria in weakly dominant strategies, economic theory has considered them ideal for guaranteeing incentive compatibility. In practice, however, experimental results document persistent deviations from dominant strategies (Kagel et al., 1987; Kagel and Levin, 1993; Chen and Sönmez, 2006; Cason et al., 2006; Healy, 2006). Consequently, recent developments in economic theory propose refinements to dominant strategies designed to make these mechanisms more effective. Various experiments have supported the principles behind these refinements in a variety of strategic environments (Esponda and Vespa, 2014, 2024; Li, 2017; Martínez-Marquina et al., 2019; Martin and Muñoz-Rodriguez, 2022; Calford and Cason, 2024; Chakraborty and Kendall, 2025).

However, comparability across these results is impossible; there is no standard on feedback from repeated rounds of play, no standard game, nor common state or strategy space. Further, comparison of these alternatives to a baseline mechanism may involve changes independent of the theoretical properties being tested. Results supporting a theoretical property may instead be due to these methodological changes rather than the simplicity property itself (e.g., Breitmoser and Schweighofer-Kodritsch, 2022).

This paper utilizes a standardized and highly relevant format to test these three simplicity properties: an initial round of value elicitation in the Cason and Plott (2014) environment. A subject makes an offer to sell an induced-value card against an unknown, randomly determined buying price. A transaction occurs at the buying price if the subject's offer is equal to or lower than this price; otherwise, no transaction takes place. The environment is particularly challenging: fewer than 20% of subjects accurately reveal their induced values upon initial play under the Becker-DeGroot-Marschak (BDM) mechanism (Cason and Plott, 2014).[1] Three proposed refinements are applicable to this environment. They are designed to generate higher rates of dominant strategy play by simplifying (i) the payoff function (Martin and Muñoz-Rodriguez, 2022), (ii) the obviousness of the mechanism's dominant strategy (Li, 2017), and (iii) the structure of the mechanism (Chakraborty and Kendall, 2025).

[1] A growing literature shows that subjects often misunderstand the BDM and fail to play its dominant strategy (Bull et al., 2019; Drichoutis and Nayga Jr, 2022; Mamadehussene and Sguera, 2023).

Though not explicitly declared in any of these three works, the very premise of these refinements suggests that simplicity should make the dominant strategy more apparent from the initial description of the problem. Each is defined in terms of what a subject can deduce before play begins, not from repeated experience. Thus, our main focus is on initial play. (We are also separately able to examine learning from experience or feedback effects from our second round of data.) To further isolate the effect of the simplicity property being tested, we utilize a 2×3 between-subjects design, comparing each behaviorally refined mechanism to an "unrefined" counterpart within the same mechanism format. We employ the BDM twice: one baseline mechanism designed to match the original design of Cason and Plott (2014), and one incorporating the behavioral refinement contingency-by-contingency (CBC). Two additional methods (a descending-price clock and a dynamic descending-price list) allow like-for-like comparisons with two other refinements, obvious strategy-proofness (OSP) and game-structure obviousness (GSO), respectively.

Martin and Muñoz-Rodriguez (2022) introduce the CBC mechanism, which has an identical structure to the standard BDM but frames each possible buying price as a potential bid by a computerized player, with each player equally likely to be selected. Building on results from several elicitation environments (specifically, Coffman, 2014; Esponda and Vespa, 2014; Möbius et al., 2022), they hypothesize that this approach reduces the perception costs associated with understanding the mechanism's payoff function. Their online experiments support this assertion, with the optimal bidding rate doubling from 17% to 34% under BDM (CBC) relative to the standard BDM. We conduct the same comparison between the BDM (CBC) and BDM (NOT CBC) in the laboratory, but with a considerably larger strategy and state space (i.e., bidding range).

Li (2017) proposes the OSP refinement, providing experimental evidence that clock-based mechanisms with equilibria in obviously dominant strategies increase the frequency of dominant strategy play. In our two CLOCK mechanisms, subjects decide when to stop a descending-price clock, indicating their minimum sell price. In the CLOCK (OSP), but not the CLOCK (NOT OSP) mechanism, the clock also stops at the drawn price. Thus, stopping at the minimum sell price is obviously dominant in CLOCK (OSP), since the only possible remaining contingencies involve an unknown price below this value. That is not the case with CLOCK (NOT OSP), where the clock does not eliminate any contingencies related to the unknown price.[2]

[2]This comparison is inspired by Breitmoser and Schweighofer-Kodritsch (2022) who use it in four player auctions

Finally, Chakraborty and Kendall (2025) propose a criterion of game-structure obviousness (GSO). A mechanism is defined as GSO if the game's structure leads to unique play of the dominant strategy, regardless of which payoff-relevant contingency subjects consider. They introduce a dynamic-price list (PL (GSO) in our experiments) that satisfies GSO, and show in experiments it achieves higher rates of dominant strategy play than the BDM. To satisfy GSO, the PL (GSO) must be descending and terminate at the unknown price, much like the CLOCK (OSP). To test the property of GSO, we compare this mechanism's performance to an uncensored price list, PL (NOT GSO), in which subjects indicate every binary choice for every contingency.[3]

To provide a comparable theoretical framework, we generalize Chakraborty and Kendall (2025) to all mechanisms. Based on the results of the aforementioned literature, we predict that each of the BDM (CBC), CLOCK (OSP), and PL (GSO) will outperform their respective unrefined counterpart.

Our baseline BDM (NOT CBC) leaves considerable room for such predictions to play out. Consistent with Cason and Plott (2014), only 26% of subjects make optimal bids in the initial round of play. The CLOCK (NOT OSP) generates even fewer optimal bids. Strikingly, the PL (NOT GSO) significantly outperforms both. Nearly half of subjects in the PL (NOT GSO) sessions bid optimally in first round play.

However, none of the behaviorally-refined mechanisms provide significantly higher rates of optimal bids than their unrefined counterparts. The BDM (CBC) does not differ from the BDM (NOT CBC) either in optimal bid rate or in the absolute deviation from the optimal bid. The CLOCK (OSP) generates similar results to the CLOCK (NOT OSP), suggesting that obviously dominant strategies do not improve the accuracy of elicitation in our environment upon initial play. While roughly half of subjects under the PL (GSO) play optimal bids (significantly outperforming the BDM), the results for the PL (NOT GSO) are very similar. It appears that the price list method, rather than the GSO property, is responsible for this result.

There is one concern before fully endorsing the price list method for its simplicity of elicitation. In terms of the absolute deviation from the optimal bid, the PL-based mechanisms do not significantly differ from the BDM in initial play. To explain this incongruous finding, we characterize

over multiple rounds.

[3]Price lists, however, also suffer from deviations from the optimal strategy of truth-telling, despite also being strategy-proof (Andersen et al., 2006, 2007; Yu et al., 2021).

the distribution of suboptimal bids across all mechanisms. We find that BDM-based mechanisms appear to be bounded largely by the first-price auction solution, consistent with previous models of game form misconception (Cason and Plott, 2014; Martin and Muñoz-Rodriguez, 2022). With CLOCK- or PL-based mechanisms, suboptimal bids instead congregate near the maximum bid. By the second round of play this distinction is irrelevant. PL-based mechanisms greatly outperform all other mechanisms on both metrics. Thus it is not only on simplicity of initial description where PL-based mechanisms excel; they are also superior at facilitating learning after one round of feedback. By round 2, 60–75% of subjects are making the optimal bid with PL-based mechanisms, double the rates of any other mechanism.

Our paper provides new, informative results that help better focus the literature around the three studied simplicity requirements. The BDM (CBC) mechanism is intended to simplify the cognitive costs associated with identifying the optimal strategy in the standard BDM (Martin and Muñoz-Rodriguez, 2022). Reconciling our results with theirs, it appears this technique is only effective when there are a small number of possible bids and states (≤ 7). While the specific technique is ineffective with a greater number, it does not mean the general idea of getting a person to consider each contingency in isolation should be dismissed (see also Niederle and Vespa, 2023). Indeed, the PL-based mechanisms may very well succeed because they make subjects go row-by-row over each contingency, suggesting the underlying principle of considering contingencies in isolation may be sound, even if the specific CBC implementation does not scale to larger state spaces.

Similarly, the OSP property (Li, 2017) and the more general clock form itself (Breitmoser and Schweighofer-Kodritsch, 2022) are found to improve bidding in multiplayer auctions relative to the sealed-bid format. Whether these refinements improve value elicitation from initial description alone (absent feedback and other bidders) was an open question prior to this work. We provide the first direct test isolating the OSP property within the clock format, comparing OSP and non-OSP clocks in a matched design. In our single-player environment, neither the OSP property nor the clock format improves optimal bidding upon initial play.

Finally, we confirm—and in some respects strengthen—the finding that the PL (GSO) mechanism generates higher rates of optimal bidding than the BDM (NOT CBC)(see Chakraborty and Kendall, 2025). However, we note this finding appears to be due to the PL structure of the PL (GSO) mechanism rather than the property of GSO. We also note that suboptimal bids under

	BDM-based	CLOCK-based	PL-based
Baseline	BDM (NOT CBC)	CLOCK (NOT OSP)	PL (NOT GSO)
Refinement	BDM (CBC)	CLOCK (OSP)	PL (GSO)

Table 1: Description of our 2×3 between-subjects treatments.

the PL (GSO) are more likely to be tied to the maximal bid than the first-price solution, which may mean both mechanisms perform similarly in another metric, the absolute deviation from the optimal bid.

Overall, our paper contributes to the literature by comparing several elicitation mechanisms that differ in dynamic structure and in whether they satisfy refinements such as CBC, OSP and GSO. We are the first to test all three in a standardized strategic environment that isolates the effectiveness of the behavioral property from the idiosyncrasies of the mechanism format. We focus on initial play, testing how well these properties make incentives simpler from description. Our results point to performance gains from the PL format rather than any theoretical refinement, which suggests future work on refinements should focus on the theoretical properties of the PL.

2 Experimental Design and Procedures

The main experiment utilized a 2×3 between-subjects design, comparing six different strategy-proof elicitation mechanisms across treatments. Table 1 summarizes the treatment variations. Three mechanisms serve as benchmark mechanisms: the BDM (NOT CBC), a descending-price clock that does not stop on the unknown price (CLOCK (NOT OSP)), and a descending-price list (PL (NOT GSO)). The other three mechanisms incorporate behavioral refinements to test their impact on the accuracy of elicitation: the contingency-by-contingency mechanism (BDM (CBC)), a standard descending clock which satisfies OSP (CLOCK (OSP)), and the game-structure obvious mechanism (PL (GSO)).

All mechanisms featured the same general structure. In each round, a decision card was distributed to subjects (see Appendix A.2 for cards). Subjects were informed that should they not sell their card they would exchange it for \$2 at the end of the experiment. Subjects were then informed of the rule by which they could keep the card or sell the card to the computer for an unknown price. Depending on the mechanism, subjects provided responses into a computer interface which

generated the minimum value for which they would sell their card. Should their minimum value exceed the unknown price, they would keep their card, otherwise they would sell it for the unknown price. Therefore, in all cases, it is weakly dominant to report a minimum selling price of \$2.

Following Cason and Plott (2014), the maximum price range $\bar{b}$ has five possible values: \$4, \$5, \$6, \$7, and \$8. We drew these values from a uniform distribution and blocked them across treatments so that each mechanism has virtually identical numbers of $\bar{b}$. The unknown price, π, was pre-determined and drawn from a range of \$0.00 to \$$\bar{b}$.00 in uniform increments of 25 cents. To ensure subjects understood that their decisions could not alter the unknown price, they were informed that it had been pre-determined, written on the back of their decision card, and covered with tape. After the computer revealed the unknown price, all subjects removed the tape and verified that it matched what had been written on their card.

2.1 Baseline Mechanisms

We first describe the three baseline mechanisms in our experiment. The BDM (NOT CBC) replicates the experimental design of Cason and Plott (2014), but in a controlled, computerized, laboratory setting. Subjects were asked to submit their "bid," which is the lowest amount at which they would be willing to sell their decision card. After subjects had submitted their bids, they were instructed to turn over their decision cards to reveal the unknown price. The computer screen would also display the unknown price and summarize the results of the round.

The CLOCK (NOT OSP) functioned in many ways similar to a standard descending-price clock. Starting from $\bar{b}$, the price would count down in 25-cent increments to zero. Participants could also choose to stop the countdown at any point. The value at which the clock stopped is called a subject's "bid," that is, the lowest price at which a subject was willing to sell their card. After the clock stopped, subjects were able to turn over their decision card and see the unknown price. Unlike a standard clock, the descending-price clock only stopped when the subject decided to stop the clock. That is, subjects did not know whether the clock had reached the unknown price until after stopping the clock.

The PL (NOT GSO) had an identical set of descending prices as the CLOCK (NOT OSP). Subjects made several binary choices one-at-a-time, deciding whether or not to sell their card at a given value. The values started at $\bar{b}$, and decreased by 25 cents at a time, down to zero. There was

no requirement that the subject should exhibit any type of consistency in these choices (i.e., single-crossing). After the final choice, subjects turned over their decision card and saw the unknown price.[4] The binary choice made at the unknown price was used to determine whether subjects would keep their card or sell it.

2.2 Mechanisms with Behavioral Refinements

Three additional mechanisms each incorporate a behavioral refinement to a baseline elicitation mechanism. The BDM (CBC) follows a similar procedure to the BDM (NOT CBC). The only difference is that each unknown price is framed as a separate computerized player under the BDM (CBC).[5] Subjects were informed in instructions that one of the computerized players would be selected and that player's price would be compared with the subject's bid. Subjects were also provided with tables that summarized all possible payoffs at each unknown price. Following the experimental design of Martin and Muñoz-Rodriguez (2022), these payoff tables present the entire payoff structure all at once, which mechanically enumerates all contingencies. Hence, the CBC mechanism is predicted to reduce the perception cost of subjects under the BDM (NOT CBC), leading to easier-to-understand payoff structures (Martin and Muñoz-Rodriguez, 2022).

The CLOCK (OSP) is designed to test the effectiveness of OSP mechanisms, and it is nearly identical to the CLOCK (NOT OSP). What distinguishes the CLOCK (OSP) is that the clock stops either when the subject stops it or when it reaches the unknown price (whichever comes first) in contrast to the CLOCK (NOT OSP), where the subject must stop the clock.

The PL (GSO) closely resembles the PL (NOT GSO). The only difference is that the countdown of binary choices in the PL (GSO) ends at the unknown price, rather than zero in PL (NOT GSO) sessions. That means subjects cannot make choices at all possible prices, following the setup of Chakraborty and Kendall (2025)'s online experiment.

[4] As introducing the word "bid" is inconsistent with the framework of a price list, a separate decision card (available in Appendix A.2) is used for the two PL-based mechanisms.

[5] There are 17 computerized players when $\bar{b}$=4, 21 computerized players when $\bar{b}$=5, 25 computerized players when $\bar{b}$=6, 29 computerized players when $\bar{b}$=7 and 33 computerized players when $\bar{b}$=8.

2.3 Experimental Procedures

Subjects sat at computer terminals and made decisions using oTree software (Chen et al., 2016). Following Cason and Plott (2014), subjects faced two rounds. The second round was announced only after subjects observed the results of the first round. In the second round, new decision cards were issued, identical to those used in the first round, except for their color. Thus, most subjects encountered a different unknown price and maximum range across the two rounds. After completing both rounds, subjects filled out a questionnaire that featured an open-ended question about their strategy used in each round, a Cognitive Reflection Test (CRT, Frederick, 2005), and an unincentivized time preference elicitation task (Coller and Williams, 1999).

At the end of the experiment, subjects were paid privately based on their totals for both rounds, plus a $10 participation payment. The average payment is $15.79. Each session took approximately 30 minutes, including instructions and payment procedures.[6]

Subjects were 386 undergraduate students at Texas A&M University, recruited using the econ-dollars.tamu.edu website, a server based on ORSEE (Greiner, 2015). Forty-one sessions took place at the Economic Research Laboratory at Texas A&M University during three time periods, January to March 2022, October to December 2022, and June 2024. 65 subjects participated in the BDM (NOT CBC) treatment (7 sessions total), 65 subjects participated in the BDM (CBC) treatment (7 sessions total), 65 subjects participated in the CLOCK (NOT OSP) treatment (8 sessions total), 65 subjects participated in the CLOCK (OSP) treatment (7 sessions total), 62 subjects participated in the PL (NOT GSO) treatment (6 sessions total), and 64 subjects participated in the PL (GSO) treatment (6 sessions total).[7]

[6]Other experiments of this type use comprehension checks, with or without feedback on correct answers. As our primary focus is on the effect of these simplicity refinements on making optimal strategy apparent from initial description, we did not include such checks. For similar reasons, comprehension checks were not included in Cason and Plott (2014)'s original experiment. It should also be noted that the direct effect of comprehension checks on subjects' performance in elicitation mechanisms remains mixed. While some experimental studies have found that comprehension checks improve subjects' understanding of elicitation mechanisms (Burchardi et al., 2021), others have provided evidence that experimental data without comprehension checks yield higher proportions of truth-telling in strategy-proof elicitation mechanisms (Chakraborty and Kendall, 2025). The impacts of additional practice rounds on subjects' comprehension are also mixed in some earlier investigations (Plott and Zeiler, 2005; Brown and Cohen, 2015; Fehr et al., 2015).

[7]To address a concern regarding the framing of our CLOCK mechanisms, an additional 62 subjects participated in another treatment with a visualized, descending clock (CLOCK (VISUAL)). The CLOCK (VISUAL) treatment (7 sessions total) studies how the framing of a descending-price clock affects subjects' type revelation in elicitation mechanisms. Subjects were asked to submit their "bid," the lowest amount that the subject would be willing to sell their decision card. In this way, the treatment begins identically to the BDM (NOT CBC). However, once subjects submitted a "bid," they would watch the visualization of a descending clock resolve the bidding. Starting from $\bar{b}$, the

3 Theoretical Framework and Predictions

We adopt the theoretical framework of Chakraborty and Kendall (2025) to incorporate all the elicitation mechanisms we study. Each player has a type drawn from the set $\mathcal{M} = \{1, 2, \ldots, n\}$ linked one-to-one to the set for valuations $\mathcal{V} = \{v^1, v^2, \ldots, v^n\}$, that is type $i \in \mathcal{M}$ has valuation v^i. Define an extensive-form game, G, with consequences in X as a tuple $(N, H, \succ, \mathcal{A}, P, (\mathcal{I}_i, u_i, z_i)_{i \in N})$, where N is the set of players (including Nature) and H is a set of nodes (histories). P is a function that maps non-terminal histories to the player who moves at that history. $(\mathcal{I}_i)_{i \in N}$ partitions the set of histories into information sets for the players and z_i maps terminal nodes to outcomes for player i. $\mathcal{A}$ maps information sets to available actions and u_i maps i's outcomes and type to a utility.

Our focus, as theirs, is on *binary allocation mechanisms* in which a single player either receives a single item or does not. A player has private valuation $v^k \in V^k = \{0.25k : k \in \mathbb{N}, 0 \leq k \leq 32\}$ which is largely consistent with their usage. A strategy, $S_1 \subset \mathcal{S}_1$, for Player 1 is a function that maps each of the player's information sets to actions available in that information set. Let $\mathcal{S}_{-1}$ be the set of all possible strategies of Nature.

Chakraborty and Kendall (2025) (Definition 1) note that for each of the mechanisms found in our experiment, each player with type $i \in \mathcal{M}$ has a unique weakly dominant strategy, S_1^i, that identifies each player's type for some realizations of S_{-1}. This means at the initial node, $h_\emptyset$, for any valuation v^k we have

$$u_1(h_\emptyset, S_1^k, S_{-1}, v^k) \geq u_1(h_\emptyset, S_1^j, S_{-1}, v^k) \quad \text{for all} \quad S_1^j \in \mathcal{S}_1, S_{-1} \in \mathcal{S}_{-1}. \tag{1}$$

We induce players to have type $k = 8$ with valuation $v^k = 2$, thus we focus on the weakly dominant strategy S_1^8 which we refer to as the "optimal bid."

Definition 1 (Optimal bid). *For BDM (NOT CBC)and BDM (CBC), the weakly dominant strategy is to bid \$2. For the CLOCK (NOT OSP), CLOCK (OSP) PL (NOT GSO) and PL (GSO) the weakly dominant strategy is to sell the card for all prices above or equal to \$2, and keep the card for all prices below or equal to \$2. All strategies uniquely identify type under some S_{-1}.*

clock would countdown in 25 cent increments until it reaches the submitted "bid" or the unknown price, whichever is higher. After the clock stops, subjects would be able to turn over their decision card and see the unknown price. Given the similarity in results between the CLOCK-based and BDM-based treatments, studying the incremental differences between these methods is unnecessary.

For simplicity, we refer to all such strategies as the "optimal bid."

The term "optimal bid" describes the above frames as the main outcome variable that we examine in our experimental data. Cason and Plott (2014) show that a majority of subjects do not provide the optimal bid under the standard BDM. We replicate the procedures of their classroom experiment in a laboratory environment with a similar student population of subjects. We expect similar results with our baseline BDM (NOT CBC) mechanism.

Our first refinement originates from the contingent thinking of the player. A triplet, (h, S_1, S_{-1}) is formally defined as a *contingency*, which is a joint conjecture about history, others' future actions, and own future actions. This definition is in line with the definition of "contingency" in Martin and Muñoz-Rodriguez (2022). The BDM (CBC) provides each computerized player's unknown price; the payoff when the seller's price is lower than or equal to the unknown price; and the payoff when the seller's price is above. All appear in the same row. As a result, each row of the payoff table uniquely corresponds to each triplet. Martin and Muñoz-Rodriguez (2022) propose that stating all possible outcomes (including possible allocations and possible states) simplifies the payoff function of the BDM, making optimal bidding under the BDM (CBC) easier and more frequent. Their experimental results confirm this prediction, albeit with a considerably smaller strategy and state space ($k < 7$). Thus, we hypothesize that the BDM (CBC) generates higher rates of optimal bids and less absolute deviation of bids than the BDM (NOT CBC).

Prediction 1. *The BDM (CBC) outperforms the BDM (NOT CBC) on both the rate of optimal bids and absolute deviation from optimal bids.*

Our second refinement focuses on simplification of the dominant strategy. Li (2017) proposes the concept of obviously dominant strategies to rationalize empirical observations that ascending clock auctions generate a higher proportion of optimal bids than second-price sealed-bid auctions. For Player 1, S_1 is defined as obviously dominant if for any alternative strategy S_1' that deviates from S_1 starting from the earliest information set, the worst possible outcome from S_1 is at least as good as the best possible outcome from S_1'. A mechanism is defined as obviously strategy-proof (OSP) if it contains an equilibrium in obviously dominant strategies.[8] Selling the card at \$2 is an obviously dominant strategy only in the CLOCK (OSP). The intuition is that for any current

[8]Formal definitions of obviously dominant strategies and OSP mechanisms are available in Li (2017).

price that is above or equal to \$2, the worst possible payoff of waiting is \$2, while stopping the clock yields a sure payoff of \$2. For any current price that is below \$2, the best possible payoff of letting the clock descend is less than \$2, while the worst possible payoff of stopping the clock at the current price is \$2. We hypothesize that the CLOCK (OSP) generates higher rates of optimal bids and less absolute deviation of bids than the CLOCK (NOT OSP).

Prediction 2. *The CLOCK (OSP) outperforms the CLOCK (NOT OSP) on both the rate of optimal bids and absolute deviation from optimal bids.*

Our third refinement places weight on simplifying the structure of the game. We borrow the following three definitions from Chakraborty and Kendall (2025). Denote $\mathcal{O}_1^G(h, S_1, S_{-1}) = \cup_{a \in \mathcal{A}^G(h)} z_1^G(h, aS_1, S_{-1})$ as the set of all possible outcomes for the player in G from the actions available at $h \in I_1$, fixing one's own future actions, S_1, and nature's future actions, S_{-1}.

Definition 2 (**Simplified structure of elicitation mechanisms**). *An extensive form game G' is a simplified structure of the original game G if G' replaces the agent's information partition by a coarser information partition, while $\mathcal{I}_{-1}^{G'} = \mathcal{I}_{-1}^{G}$ and $\mathcal{S}_{-1}^{G'} = \mathcal{S}_{-1}^{G}$.*

Definition 3 (**Payoff-relevant contingency and payoff-relevant information set**). *A contingency is payoff-relevant at I_1 if $h \in I_1$ and $\left|\mathcal{O}_1^G(h, S_1, S_{-1})\right| > 1$. An information set I_1 is called a payoff-relevant information set if it has at least one payoff-relevant contingency.*

Definition 4 (**Uniquely justifiable strategies**). *An action $a \in \mathcal{A}(I_1)$ is uniquely justifiable if it is the only action that maximizes the agent's utility regardless of the payoff-relevant contingency at I_1. A strategy S_1 is a uniquely justifiable strategy if every $S_1(I_1)$ is a uniquely justifiable action at every payoff-relevant information set I_1.*

Chakraborty and Kendall (2025) define a mechanism, G, as *game-structure obvious* (GSO) if (1) it induces Player 1 to reveal type (play S^k) in weakly dominant strategies; (2) has some realizations of S_{-1} that can uniquely separate types; (3) most crucially, there exists a simplified game G' of G such that playing the strategy consistent with type is uniquely justifiable.

All mechanisms studied here satisfy (1) and (2) as type revelation in weakly dominant strategies is a fundamental principle of strategy-proof elicitation mechanisms. But the PL (GSO) is the only mechanism studied to satisfy criterion (3). The reason is that the PL (GSO) has G' with coarser

information sets, represented with a separate contingency for every realization of the drawn price. At any information set I_1, all possible actions are either choosing to sell the card or keep the card at a specific price. All possible contingencies, (h, S_1, S_{-1}), can be categorized by whether this specific unknown price will be drawn or not. Hence, for information set I_1 corresponding to a drawn price above (below) \$2 selling (keeping) the card is the only uniquely justifiable action. Thus, there is a unique payoff-relevant contingency at each information set (the clock price), and all uniquely justifiable actions at every payoff-relevant information set form the dominant strategy in the PL (GSO): Selling the card at every clock price that is greater than or equal to \$2, while keeping the card otherwise.

Static mechanisms do not have this property ($G = G'$) as they have only one information set. Bidding \$2 is not a uniquely justifiable action at the initial node. There are contingencies where the unknown price is greater than or equal to \$2, then submitting a bid that is below the unknown price is also justifiable. Similarly, for any contingency where the unknown price is less than \$2, submitting a bid that is greater than or equal to \$2 is justifiable. While the CLOCK (OSP) is uniquely different because it has multiple information sets, it does not satisfy GSO either. Consider the information set where the clock has fallen to a price of \$3. Selling the card (i.e., not stopping the clock) is not a uniquely justifiable action. Consider the contingency where the seller chooses to let the clock fall to zero and the unknown price is also zero. Stopping the clock at \$3 is better for the seller than letting the clock continue for this specific contingency.

The only mechanism that satisfies GSO is PL (GSO), which is otherwise very similar to PL (NOT GSO). Both mechanisms have similar weakly dominant strategies for Player 1 that lead her to reveal her type over binary choices on a price list. However, PL (NOT GSO) does not satisfy GSO because it is intrinsically static. We would expect better performance with the PL (GSO).

Prediction 3. *The PL (GSO) outperforms the PL (NOT GSO) on both the rate of optimal bids and absolute deviation from optimal bids.*

4 Results

We proceed in four steps: we first confirm our BDM replicates Cason and Plott (2014), then compare optimal bid rates, absolute deviations, and learning effects across all six mechanisms.

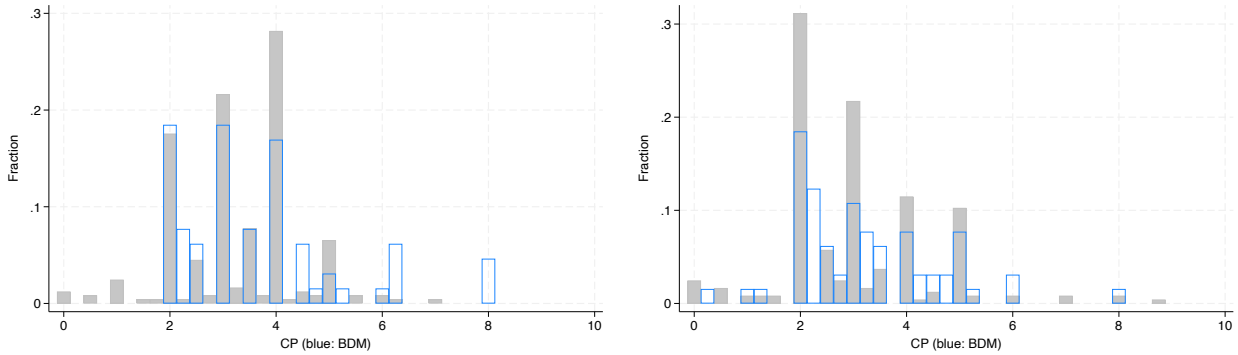

Figure 1: Histogram of bids in Cason and Plott (2014) vs. our replicated BDM (NOT CBC) in rounds 1 (a, left) and 2 (b, right).

Dependent variable: plays optimal strategy ($1.75 ≤ bid ≤ $2.25)				
Variables	round 1		round 2	
our study	0.078	0.078	-0.003	-0.004
	(0.074)	(0.075)	(0.040)	(0.039)
max bid - 6		-0.002		-0.015
		(0.015)		(0.012)
constant	0.184***	0.184***	0.310***	0.311***
	(0.015)	(0.015)	(0.021)	(0.021)
observations	310	310	309	309
r-squared	0.008	0.114	0.002	0.109

*** p<0.01, ** p<0.05, * p<0.1

Table 2: Regressions of optimal bid in Cason and Plott (2014) and our BDM (NOT CBC) mechanism, a direct replication of their work. Both regressions utilize cluster-robust standard errors at the session level.

Our BDM (NOT CBC) was designed to replicate the classroom experiment of Cason and Plott (2014) in the laboratory. We find, as they do, that the majority of subjects fail to play a dominant strategy over two rounds. Defining a subject's behavior as "optimal" if their bid is between $1.75 and $2.25, only 26.15% (17 of 65) of round 1 and 30.77% (20 of 65) of round 2 bids can be classified as optimal. The equivalent numbers in their study are 18.37% (45 of 245) and 31.02% (76 of 245), respectively (chi-square tests: $p \approx 0.163$ (round 1), $p \approx 0.969$ (round 2)). The distributions of bids also appear similar (Kolmogorov-Smirnov tests: $p \approx 0.298$ (round 1), $p \approx 0.222$ (round 2); see Figures 1(a) and 1(b)). Tables 2 and 3 provide regression results that also control for the bid range. The regressions confirm there is little difference between our BDM (NOT CBC) mechanism and Cason and Plott (2014)'s, although their settings, populations, and years differed: mean deviations are $1–2 away from the optimal bid (a relative difference of 50-100%).

The focus of this paper is to investigate whether any mechanism might improve upon this

Dependent variable: absolute deviation from optimal strategy (|bid − 2|)				
Variables	round 1		round 2	
our study	0.245	0.247	0.127	0.165
	(0.206)	(0.214)	(0.128)	(0.103)
max bid - 6		0.254***		0.283***
		(0.048)		(0.031)
constant	1.382***	1.384***	1.188***	1.181***
	(0.066)	(0.068)	(0.023)	(0.022)
observations	310	310	310	310
r-squared	0.006	0.006	0.000	0.002

*** p<0.01, ** p<0.05, * p<0.1

Table 3: Regressions of absolute deviation from optimal bid in Cason and Plott (2014) and our BDM (NOT CBC) mechanism, a direct replication of their work. Both regressions utilize cluster-robust standard errors at the session level.

BDM (NOT CBC) baseline and whether this improvement can be isolated and attributed to a corresponding simplicity refinement from the literature of behavioral mechanism design (i.e., CBC, GSO, and OSP). To begin, we must establish a standardized definition of an optimal response. As before, we can define subjects' behavior as optimal if they submit a bid that is between \$1.75 and \$2.25 under mechanisms with the BDM format. For clock-based mechanisms, i.e., CLOCK (OSP) and CLOCK (NOT OSP), we can adopt the same procedure. For PL-based mechanisms, PL (GSO) and PL (NOT GSO), we define a subject's behavior as optimal if it changes from "sell" to "keep" (the card) at a price that is between \$1.75 and \$2.25, and this specific change is the single point of switching.[9]

Both the CLOCK (OSP) and PL (GSO) have issues with censored observations. In the CLOCK (OSP), if the clock stops at the unknown price rather than by the subject's choice, we only know that a subject's "bid" is at most the value of the unknown price. Similarly, in the PL (GSO), if a subject always selects "sell," we only record a minimum selling value of at most the level of the unknown price. In the CLOCK (OSP), 9 (14%) and 21 (32%) of 65 observations per round are censored in rounds 1 and 2, respectively. In the PL (GSO), the corresponding numbers are 15 (23%) and 28 (44%) of 64 per round.

We begin with a bounds-based approach to control for the censoring of observations. Figures 2(a) and 2(b) show rates of optimal bids for each mechanism in the first and second rounds of

[9]Thus, all of the following circumstances are regarded as suboptimal under PL-based mechanisms: (1) a subject switches between "sell" and "keep" more than once; (2) a subject switches only once from "sell" to "keep," but the switching price is not between \$1.75 and \$2.25; (3) a subject starts from "keep," but switches at "sell" in later prices; (4) a subject keeps the card at all possible prices; and (5) a subject sells the card at all possible prices.

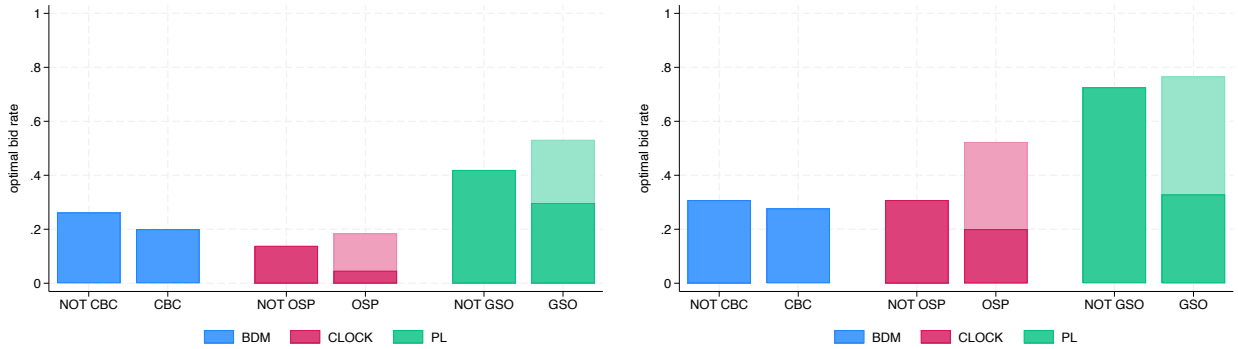

Figure 2: Rate of optimal strategic play in rounds 1 (a, left) and 2 (b, right) for all six mechanisms. The CLOCK (OSP) and PL (GSO) include observations that were consistent with playing the optimal strategy until the point of censoring. For these two mechanisms, the shaded bars indicate bounds given the uncertainty of these subjects' strategies. The lowest (highest) bar is the interpretation of none (all) of these subjects would have played the optimal strategy. (N=386 in each graph)

the experiment. For the two mechanisms with censoring, the upper bar assumes that all censored subjects would have played the optimal strategy, while the lower bar assumes that none would have done so. The true, unobserved rate must lie in the interval defined by these bounds.

The figure reveals a clear pattern: the PL format yields a substantially higher rate of optimal bids than mechanisms based on the BDM or CLOCK format, and this advantage is present in both rounds. By round 2, nearly 80% of subjects are playing an optimal strategy under the PL (NOT GSO). Even under the worst-case assumption that none of the censored observations would have yielded the optimal bid, PL (GSO) still achieves a higher rate of optimal play than the other four non-PL mechanisms.

The data provide no support for any of the three simplicity refinements improving subject play. The BDM (CBC) achieves lower rates of optimal play than BDM (NOT CBC), suggesting formatting the BDM contingency-by-contingency has no effect on subjects' comprehension. Behavioral refinements of CLOCK (OSP) and PL (GSO) do not appear to improve results relative to their non-refined counterparts, but this result is trickier to infer from the figure due to the censoring of observations.

We turn to censored regression techniques to make statistical inference here. This technique has the benefit of analyzing all of our data, but makes specific parametric assumptions about the

unobserved values.[10] We follow a standard linear probability model: denote y_i as a binary variable on whether subject $i \in N$ bids optimally ($y_i = 1$) or not ($y_i = 0$), x_i as covariates of the subject, β as corresponding coefficients, and ϵ_i as the error term that is normally distributed as $\epsilon \sim N(0, \tilde{\sigma}^2)$. Denote $\Phi()$ as the cumulative standard normal distribution.

There are two different cases in our data. Observations $i \in \mathcal{C}$ are fully-observed point data. Observations $i \in \mathcal{R}$ are right-censored, where y_i is greater than or equal to $y_{\mathcal{R}i}$, a censoring value observed from data. We are concerned about cases where the subject was censored before they could make an optimal bid (i.e., the unknown price was above \$2.25). These are coded as "0" but could have been "1" in the absence of censoring. A log-likelihood function is maximized for the observed point values and the censored values under the following specification:

$$\log L = -\frac{1}{2} \sum_{i \in \mathcal{C}} \left[\left(\frac{y_i - x_i\beta}{\tilde{\sigma}} \right)^2 + \log(2\pi\tilde{\sigma}^2) \right] + \sum_{i \in \mathcal{R}} \log \left[1 - \Phi \left(\frac{y_{\mathcal{R}i} - x_i\beta}{\tilde{\sigma}} \right) \right] \tag{2}$$

Table 4 provides results of this regression for rounds 1 and 2 with and without additional control variables. The results are similar across specification and confirm what we already suspected: none of the three simplicity refinements produce positive and significant deviations from their unrefined baseline mechanism. Compared to the baseline BDM (NOT CBC), the other three non-PL mechanisms do not produce significantly greater performance. Both PL-based mechanisms (PL (GSO) and PL (NOT GSO)) outperform the BDM (NOT CBC) as well as all other mechanisms on magnitudes of 15-40 percentage points. The effect appears stronger in round 2. We will investigate this potential learning mechanism in subsequent analysis.

We next examine the mean absolute deviation of bids, the mean absolute value of the difference between a subject's bid and \$2. This value provides a continuous measure of a subject's deviation from selling the card at the induced value. Figures 3(a) and 3(b) plot the mean value of this measure across mechanisms for rounds 1 and 2. As before, for the two mechanisms that involve censoring, the upper bar represents an interpretation that subjects would have sold at the censored value while the lower bar assumes that all censored subjects would have played the optimal strategy and thus, produce zero absolute deviation.

[10] There are several ways to proceed here. In a similar analysis, Chakraborty and Kendall (2025) addressed this problem by only looking at observations where the randomly determined unknown price was less than \$2. We like this method less because it discards roughly 60% of our data. We find similar results using this approach.

	(1)	(2)	(3)	(4)	(5)	(6)
	Dependent variable: plays optimal strategy ($1.75 ≤ bid ≤ $2.25)					
Variables	round 1			round 2		
BDM (CBC)	-0.062	-0.062	-0.029	-0.031	-0.030	0.016
	(0.082)	(0.088)	(0.075)	(0.058)	(0.054)	(0.058)
CLOCK (NOT OSP)	-0.123	-0.123	-0.122*	0.000	-0.000	-0.002
	(0.077)	(0.080)	(0.070)	(0.053)	(0.051)	(0.042)
CLOCK (OSP)	-0.165**	-0.165**	-0.148**	0.065	0.061	0.074
	(0.078)	(0.081)	(0.074)	(0.072)	(0.067)	(0.087)
PL (NOT GSO)	0.160**	0.159*	0.194**	0.332***	0.327***	0.364***
	(0.081)	(0.085)	(0.076)	(0.047)	(0.046)	(0.056)
PL (GSO)	0.158*	0.158*	0.168**	0.418***	0.427***	0.435***
	(0.081)	(0.083)	(0.079)	(0.047)	(0.044)	(0.045)
maximum bid - 6		-0.031***	-0.032***		-0.036**	-0.029*
		(0.012)	(0.011)		(0.016)	(0.015)
constant	0.262***	0.261***	0.250***	0.308***	0.304***	0.289***
	(0.043)	(0.073)	(0.061)	(0.021)	(0.021)	(0.036)
subject controls	N	N	Y	N	N	Y
observations	386	386	386	386	386	386
right-censored	24	24	24	49	49	49
log-likelihood	-196.8	-194.5	-174.4	-223.8	-221.5	-196.2

*** p<0.01, ** p<0.05, * p<0.1

Table 4: Censored regressions of optimal bid on mechanism. All regressions utilize cluster-robust standard errors at the session level. Subject demographic controls are relative to mean value and include gender, whether a subject is an econ major, and number of correct answers on CRT.

Mechanism performance is more ambiguous under this metric. Despite their lower optimal bid rates, BDM-based mechanisms appear to have the lowest absolute deviation in round 1. In round 2, the two PL-based mechanisms, specifically PL (NOT GSO), have the lowest mean absolute deviation. Consistent with the past result on optimal bid rate, the CLOCK-based mechanisms have the highest mean absolute deviation.

We again use censored regression techniques to account for censored observations (Table 5). In this case our censored observations provide an upper rather than lower bound: the value at censoring is the maximum possible absolute deviation; they are left censored. We index such observations by $i \in \mathcal{L}$ where y_i is less than or equal to $y_{\mathcal{L}i}$. The corresponding regression equation is

$$\log L = -\frac{1}{2}\sum_{i\in\mathcal{C}}\left[\left(\frac{y_i - x_i\beta}{\tilde{\sigma}}\right)^2 + \log(2\pi\tilde{\sigma}^2)\right] + \sum_{i\in\mathcal{L}}\log\Phi\left(\frac{y_{\mathcal{L}i} - x_i\beta}{\tilde{\sigma}}\right).$$

As before, we first note that there is no evidence to suggest the three behavioral refinements significantly outperform their respective baseline mechanisms. Second, in round 1, BDM-based mechanisms appear to produce statistically equal absolute deviation as PL-based mechanisms.

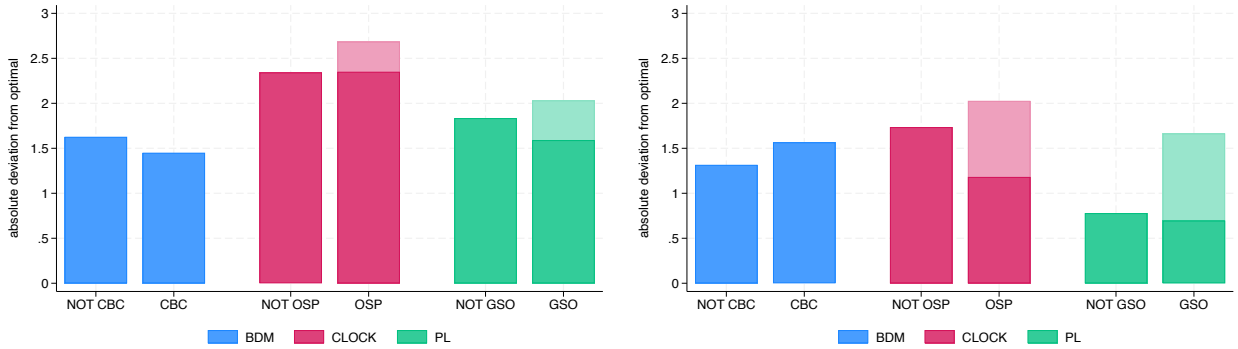

Figure 3: Absolute deviation from optimal strategy (i.e., \$2 bid) in rounds 1 (a, left) and 2 (b, right) under all six mechanisms. The CLOCK (OSP) and PL (GSO) include observations that were consistent with playing the optimal strategy until the point of censoring. For these two mechanisms, the shaded bars indicate bounds given the uncertainty of these subjects' strategies. The lowest (highest) bar denotes the case where all censored subjects would have played a strategy that is exactly (furthest from) the optimal strategy of \$2. (N=386 in each graph)

By round 2, PL-based mechanisms outperform all others by \$0.50–\$0.60 (33–45%) in absolute deviation. Both CLOCK-based mechanisms perform worse on average by roughly the same amount.

How can the PL mechanisms generate higher rationality rates in round 1 than the BDM mechanisms, yet yield comparable absolute deviations? The answer must lie in the pattern of subjects' suboptimal bids. Figure 4 visually illustrates the pattern of all bids across mechanisms and round.

Panels (a), (c) and (e) provide empirical CDFs of bids for the respective BDM, CLOCK, and PL mechanisms in round 1; Panels (b), (d), and (f) for round 2. The graphs are standardized to include four reference points: 0, the item's value v fixed at \$2, the maximum bid $\hat{b} \in \{\$4, \$5, \$6, \$7, \$8\}$, and the midpoint between the item's value and the maximum bid, $\frac{v+\hat{b}}{2} \in \{\$3, \$3.50, \$4, \$4.50, \$5\}$. Note that the final value would be the optimal bid if the BDM were actually a first-price auction. As noted earlier, Cason and Plott (2014) suggest that suboptimal bids in the BDM mechanism are due to a specific game form misconception where subjects believe they are playing a first-price auction. Martin and Muñoz-Rodriguez (2022) extend this idea and develop a model that uses rational inattention (Sims, 2003) to predict bids in the BDM will be a convex combination of the optimal bids in a second and first-price auction. The model implies all bids should fall on $[v, \frac{v+\hat{b}}{2}]$. They find empirical support in both their data and Cason and Plott (2014).

Looking at Figures 4(a) and (b), we see this intuition largely holds for the two BDM-based mechanisms, BDM (NOT CBC) and BDM (CBC). In either round, roughly $\frac{2}{3}$ of suboptimal bids

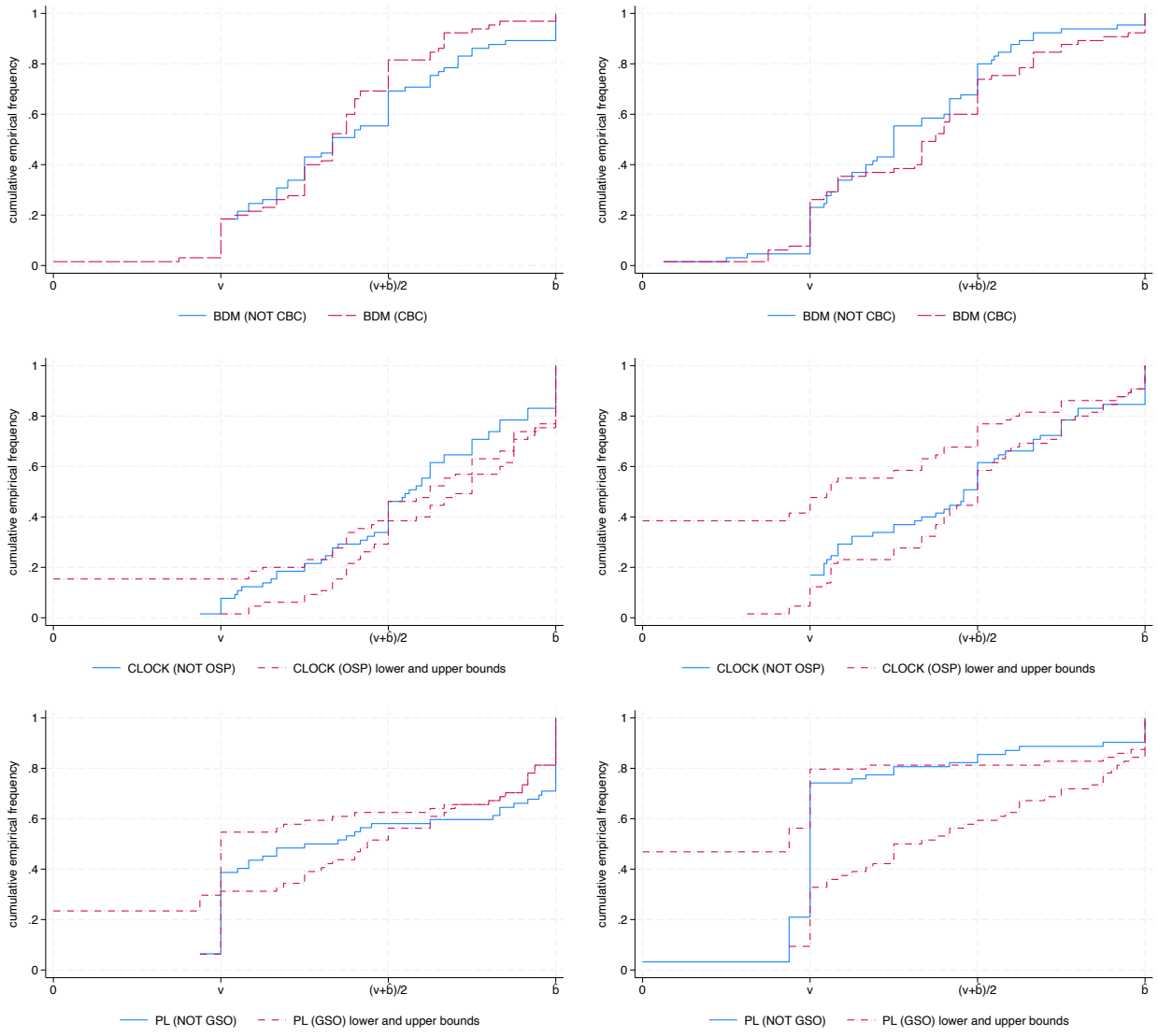

Figure 4: Empirical CDF of bids by mechanism and round, standardized by item value (v) and maximum possible bid ($\hat{b}$). Top row: BDM (NOT CBC) and BDM (CBC) in round 1 (a, left) and 2 (b, right). Middle row: CLOCK (NOT OSP) and CLOCK (OSP) in round 1 (c, left) and 2 (d, right). Bottom row: PL (NOT GSO) and PL (GSO) in round 1 (e, left) and 2 (f, right). For the mechanisms that involve censored bids, CLOCK (OSP) and PL (GSO), upper and lower bounds of the empirical CDF are provided under the assumptions that every censored bid is 0 and that every censored bid equals the censoring point, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
	Dependent variable: absolute deviation from optimal strategy ($\lvert \text{bid} - 2 \rvert$)					
Variables		round 1			round 2	
BDM (CBC)	-0.177	-0.177	-0.295	0.250	0.244*	0.120
	(0.258)	(0.276)	(0.231)	(0.178)	(0.131)	(0.132)
CLOCK (NOT OSP)	0.715**	0.715**	0.703**	0.419*	0.419**	0.412**
	(0.321)	(0.297)	(0.278)	(0.216)	(0.206)	(0.177)
CLOCK (OSP)	0.852***	0.874***	0.828***	0.088	0.176	0.160
	(0.302)	(0.287)	(0.283)	(0.248)	(0.208)	(0.228)
PL (NOT GSO)	0.208	0.199	0.170	-0.537***	-0.629***	-0.648***
	(0.259)	(0.258)	(0.244)	(0.201)	(0.161)	(0.129)
PL (GSO)	0.050	0.097	-0.003	-0.554***	-0.483***	-0.553***
	(0.292)	(0.299)	(0.270)	(0.210)	(0.171)	(0.181)
maximum bid - 6		0.576***	0.404***		0.362***	0.341***
		(0.055)	(0.052)		(0.055)	(0.052)
constant	1.627***	1.636***	1.674***	1.315***	1.355***	1.392***
	(0.187)	(0.195)	(0.121)	(0.042)	(0.037)	(0.037)
subject controls	N	N	Y	N	N	Y
observations	386	386	386	386	386	386
left-censored	24	24	24	49	49	49
log-likelihood	-725.1	-673.3	-652.4	-632.3	-608.8	-585.8

*** p<0.01, ** p<0.05, * p<0.1

Table 5: Censored regressions of absolute deviation from optimal bid on mechanisms. All regressions utilize cluster-robust standard errors at the session level. Subject demographic controls are relative to mean value and include gender, whether a subject is an econ major, and number of correct answers on CRT.

fall between the second-price and first-price solution.

For the CLOCK and PL mechanisms, the distributions of suboptimal bids appear different. In round 1, under each, more than half of suboptimal bids fall above the first-price optimal bid. For any possible configuration of censored bids, the difference between BDM and non-BDM mechanisms is statistically different (χ^2 p-value ≈ 0.000). The pattern is nearly the same for round 2 (χ^2 p-value $\in (0.000, 0.0704]$).

By round 2, however, the higher distribution of suboptimal bids no longer affects the performance of the PL mechanisms in terms of overall absolute deviation. Both outperform the BDM mechanisms on that metric. Indeed it appears that for a majority of subjects one round of feedback with a PL mechanism is sufficient to get them to play the optimal strategy.

There are two ways in which feedback can help. It can reinforce to subjects playing the optimal strategy that they are indeed playing the optimal strategy. It can also inform subjects playing suboptimally of their poor strategic choice. Figures 5(a) and (b) provide rates of subject optimal bidding in the second round across mechanisms conditional on first round behavior. On the left (a)

		Bids greater than optimal bid				
		round 1			round 2	
mechanism type	total	bid $\in (v, \frac{v+\hat{b}}{2}]$	bid $\in (\frac{v+\hat{b}}{2}, \hat{b}]$	total	bid $\in (v, \frac{v+\hat{b}}{2}]$	bid $\in (\frac{v+\hat{b}}{2}, \hat{b}]$
BDM	98	66	32	85	55	30
		(67%)	(33%)		(65%)	(35%)
CLOCK	74–88	39–43	52–70	75–96	35–44	40–52
		(35–45%)	(54–65%)		(40–52%)	(48–60%)
PL	56–63	13–24	50–54	28–58	7–23	21–35
		(19–32%)	(68–80%)		(17–52%)	(48–83%)
χ^2 test BDM vs. non-BDM			0.000			0.000–0.0704

Table 6: Categorization of suboptimal overbids by mechanism type, rounds 1 and 2. A bid is classified whether it falls on the interval between the optimal bid in the second-price mechanism, v, and the optimal bid in a similar first-price mechanism, $\frac{v+\hat{b}}{2}$. Alternatively it may fall on an interval above the first-price optimal bid up to the maximum bid, $\hat{b}$. A small percentage of suboptimal bids fall below the true value v. Ranges of bids are constructed to account for all possible values given censoring of CLOCK (OSP) and PL (GSO) mechanisms. χ^2 test range are taken from range of p-values of test under all such possibilities.

shows the rates at which subjects already playing an optimal strategy in round 1 continue to play it in round 2. On the right (b) shows the rate at which subjects not playing the optimal strategy in round 1 adopt it in round 2. Because the rates involve a determination of optimal strategy in both rounds, they are particularly sensitive to censoring and make classification imprecise. We nonetheless provide bounds for our censored mechanisms. To provide point estimates, we once again follow the regression framework in Equation (2) to estimate the incidence of optimal bids in round 2. However, we add an additional control term, a dummy variable on whether the round 1 bid was optimal. Censoring in round 1 for the CLOCK (OSP) and PL (GSO) presents another problem, so we replace the dummy with the predicted $\hat{y}$ variable for censored observations in round 1. Full results are found in Appendix Table A.1 and treatment effects are plotted in Figures 6 (a) and (b).

We observe that about 70% of subjects already playing the optimal strategy in round 1 under the BDM-based mechanisms continue to do so in round 2. While this rate may seem high, we note under the PL-based mechanisms the rate is 20 percentage points higher: over 90% of subjects will continue to make the optimal bid in round 2 if they made it in round 1. The difference is more extreme for subjects that do not make the optimal bid in round 1. Only 17% of these subjects learn to make the optimal bid in round 2 under the baseline BDM (NOT CBC). With the PL (GSO)

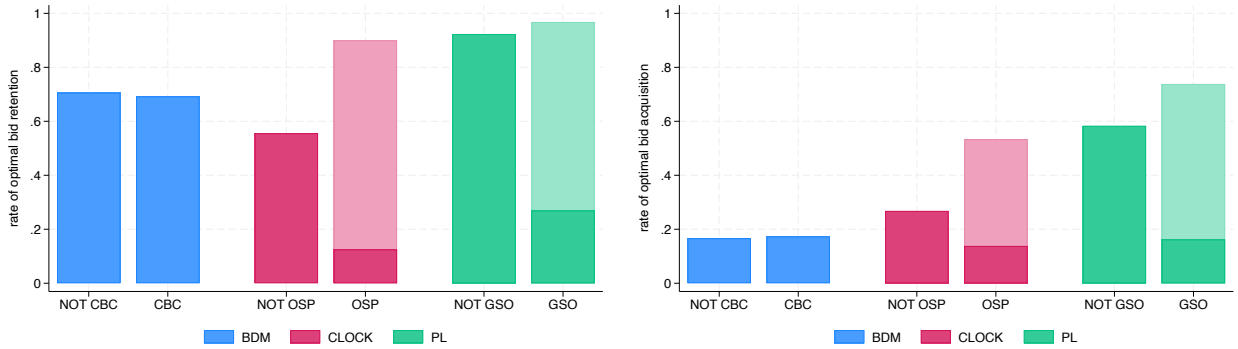

Figure 5: Rates playing the optimal bid in round 2 conditional on bid in first round across mechanisms. (a, left) Rates of making the optimal bid having previously made the optimal bid. (b, right) Rates of making optimal bid having previously not made optimal bid. For the CLOCK (OSP) and PL (GSO), the shaded bars indicate bounds on rates given the uncertainty of censored subjects' strategies. Censored round 2 observations are interpreted as in Figure 2: the lower (upper) bound assumes none (all) would have played the optimal strategy. Censored round 1 observations additionally affect whether a subject appears in panel (a) or panel (b); the bounds reflect the classification that minimizes and maximizes the conditional rate in each panel. (N=111–118, left; N=337–344, right)

and PL (NOT GSO), the rates are estimated to be 30–40 percentage points higher, suggesting that about half of subjects not making the optimal bid initially learn to play it in one round.

5 Conclusion

Over the last ten years, several behavioral refinements have been suggested to improve subject comprehension of incentives regarding dominant strategies in strategic play. We investigate whether any of these behavioral refinements generate more accurate value elicitation in initial play. We choose a specific environment where the standard BDM is known to fail for a majority of subjects initially and after a repeated round of play.

While past experimental work has supported each of these behavioral refinements in other environments, our focus is on initial play in a standardized, widely used environment with a large number of states. We wish to examine whether, from description alone, any of these simplicity refinements can make the problem easier for subjects. For the CBC and OSP refinements the answer is a clear no. The BDM (CBC) and CLOCK (OSP) do not perform significantly better in either optimal bid rate or absolute deviation of bids than the baseline BDM (NOT CBC). Past work supporting the superior performance of OSP mechanisms, or clock mechanisms in general,

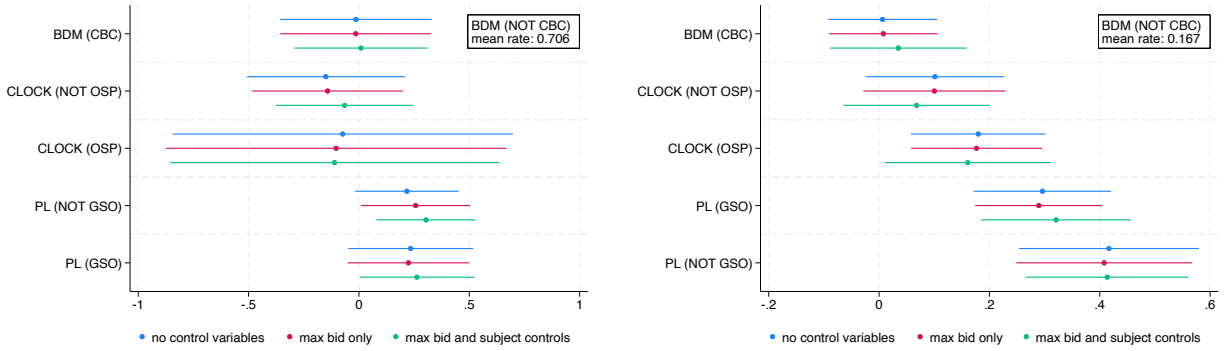

Figure 6: Point estimates from a regression model of playing an optimal bid in round 2 on mechanism (a, left) and mechanism interacted with making an optimal bid in round 1 (b, right). Right-censored values for the dependent variable are handled using maximum likelihood estimation in Equation (2). Right-censored values for the regressor, optimal strategy (round 1), are predicted from the corresponding model's regression results in Table 4. All regressions utilize cluster-robust standard errors at the session level. Three specifications include no controls, a control for the maximum bid value, and controls for the maximum bid value and subject-specific controls. Subject controls are relative to mean value and include gender, whether a subject is an econ major, and number of correct answers on CRT. Full details of the regression are available in Appendix Table A.1.

over second-price sealed-bid mechanisms has based this conclusion on multiple rounds of data in games of several players (Li, 2017; Breitmoser and Schweighofer-Kodritsch, 2022). Our results show these mechanisms do not produce similar benefits upon initial play in our single-player environment.

Past experiments also support that framing contingent states of a problem one-by-one may aid in subject comprehension. However, specific to value elicitation, Martínez-Marquina et al. (2019) and Esponda and Vespa (2024) draw this conclusion from strategic environments with only two and three possible states, respectively. Other studies that successfully utilize the CBC method with more states involve eliciting probability rather than valuation (Coffman, 2014; Möbius et al., 2022), which may be an inherently simpler task (Ngangoué and Schotter, 2023).[11] We take our BDM (CBC) mechanism directly from Martin and Muñoz-Rodriguez (2022), who examine this problem with seven possible states and find that BDM (CBC) outperforms BDM (NOT CBC) in initial play. Our results suggest that this advantage is not robust when the number of states increases to 17–33, which greatly limits the applicability of the technique. One possible explanation is that BDM (CBC) performs better only in relatively small state spaces; once the number of states

[11] Similarly, Hao and Houser (2012) find that a clock mechanism improves the accuracy of belief elicitation upon initial play, though the mechanism does not satisfy OSP.

exceeds subjects' limits of working memory (≈ 7, Miller, 1956), its advantage over BDM (NOT CBC) appears to disappear. Note, however, that the PL-method offers an alternative way to view contingencies one-by-one; subjects must consider each row individually. Unlike the CBC table, which grows with the number of states, the PL presents one binary choice at a time regardless of the state space, suggesting this structural feature is more robust to scale.

Our results do confirm that the PL (GSO) mechanism greatly increases the number of subjects that correctly play the optimal strategy on initial play compared to the BDM (NOT CBC). This result is consistent with Chakraborty and Kendall (2025), who find it over a smaller state space. However, we are able to attribute these results to the price list structure of the mechanisms, rather than their theoretical property of game-structure obviousness. The PL (NOT GSO) performs similarly well. The results suggest price lists are better than sealed bids or clocks at inducing subjects to play the optimal bid upon initial play in these environments. In general, this suggests some components of game-structure obviousness may very well be pivotal in these environments.

There is one qualification to our full endorsement of the price list method of elicitation. Our second performance metric, the absolute deviation of bids, is not significantly different between the BDM and PL-based mechanisms in round 1. This result occurs because bids in the BDM-based mechanisms appear to be bounded largely by the first-price auction solution, consistent with previous models of game form misconception (Cason and Plott, 2014; Martin and Muñoz-Rodriguez, 2022). Suboptimal bids do not follow this pattern with CLOCK- or PL-based mechanisms. They are more likely to congregate near the maximum bid. Nonetheless, in a second round of play, PL-based mechanisms greatly outperform on both metrics. Thus it appears that a price list produces more comprehension before feedback than any other method; it also produces more learning after a single round of feedback than any other method. By round 2, 60–75% of subjects are making the optimal bid with PL-based mechanisms, double the rates of any other mechanism.

Thus we are left to wonder what behavioral refinement inherent in the PL-method makes the elicitation method so simple. It does appear consistent with some of the general intuition behind each of the CBC, OSP, and GSO theories without fully satisfying the properties of any. We leave it to future research to determine which characteristics are relevant for theoretical formalization. Indeed, some work in this area is already emerging (e.g., Healy, 2020).

Declaration of Generative AI and AI-assisted Technologies in the Writing Process

During the preparation of this work, the authors used ChatGPT and Claude to proofread existing content. After using this tool/service, the authors reviewed and edited the content as needed. The authors take full responsibility for the content of the publication.

Funding Sources

This study was funded by the Economics Excellence Fund at Texas A&M University, the Dr. John Van Huyck Endowed Fellowship at Texas A&M University, the Fundamental Research Funds at Shandong University (Grant number: 11020064510004), and the Natural Science Foundation of Shandong Province (Grant number: ZR2025QC1251).

Acknowledgements

This paper's earliest draft was completed when Jinliang Liu was a Ph.D. student at Texas A&M University. We thank Yonghong An, Tim Cason, Marco Castillo, Anujit Chakraborty, Andreas Drichoutis, Catherine Eckel, Paul Feldman, Huiyi Guo, Rustamdjan Hakimov, Paul J. Healy, Chad Kendall, Dan Levin, Jianbiao Li, Shengwu Li, Daniel Martin, Eric Maskin, Marco Palma, Ragan Petrie, Charles Plott, Tim Salmon, Sebastian Schweighofer-Kodritsch, Danila Serra, Rodrigo Velez, Emanuel Vespa, Lise Vesterlund, and participants from various seminars and conferences for helpful comments and discussions. This study was approved by the Institutional Review Board (IRB) at Texas A&M University (IRB2021-1551M). All errors are our own. Declaration of interest: none.

References

Andersen, S., Harrison, G. W., Lau, M. I., and Rutström, E. E. (2006). Elicitation using multiple price list formats. *Experimental Economics*, 9:383–405.

Andersen, S., Harrison, G. W., Lau, M. I., and Rutström, E. E. (2007). Valuation using multiple price list formats. *Applied Economics*, 39(6):675–682.

Breitmoser, Y. and Schweighofer-Kodritsch, S. (2022). Obviousness around the clock. *Experimental Economics*, 25(2):483–513.

Brown, A. L. and Cohen, G. (2015). Does anonymity affect the willingness to accept and willingness to pay gap? A generalization of Plott and Zeiler. *Experimental Economics*, 18(2):173–184.

Bull, C., Courty, P., Doyon, M., and Rondeau, D. (2019). Failure of the Becker–DeGroot–Marschak mechanism in inexperienced subjects: New tests of the game form misconception hypothesis. *Journal of Economic Behavior & Organization*, 159:235–253.

Burchardi, K., De Quidt, J., Gulesci, S., Lerva, B., and Tripodi, S. (2021). Testing willingness to pay elicitation mechanisms in the field: Evidence from Uganda. *Journal of Development Economics*, 152:102701.

Calford, E. M. and Cason, T. N. (2024). Contingent reasoning and dynamic public goods provision. *American Economic Journal: Microeconomics*, 16(2):236–266.

Cason, T. N. and Plott, C. R. (2014). Misconceptions and game form recognition: Challenges to theories of revealed preference and framing. *Journal of Political Economy*, 122(6):1235–1270.

Cason, T. N., Saijo, T., Sjöström, T., and Yamato, T. (2006). Secure implementation experiments: Do strategy-proof mechanisms really work? *Games and Economic Behavior*, 57(2):206–235.

Chakraborty, A. and Kendall, C. (2025). Uniquely justifiable strategies and mechanism design with boundedly rational players. *Working Paper*.

Chen, D. L., Schonger, M., and Wickens, C. (2016). oTree—An open-source platform for laboratory, online, and field experiments. *Journal of Behavioral and Experimental Finance*, 9:88–97.

Chen, Y. and Sönmez, T. (2006). School choice: An experimental study. *Journal of Economic Theory*, 127(1):202–231.

Coffman, K. B. (2014). Evidence on self-stereotyping and the contribution of ideas. *The Quarterly Journal of Economics*, 129(4):1625–1660.

Coller, M. and Williams, M. B. (1999). Eliciting individual discount rates. *Experimental Economics*, 2:107–127.

Drichoutis, A. C. and Nayga Jr, R. M. (2022). Game form recognition in preference elicitation, cognitive abilities, and cognitive load. *Journal of Economic Behavior & Organization*, 193:49–65.

Esponda, I. and Vespa, E. (2014). Hypothetical thinking and information extraction in the laboratory. *American Economic Journal: Microeconomics*, 6(4):180–202.

Esponda, I. and Vespa, E. (2024). Contingent thinking and the sure-thing principle: Revisiting classic anomalies in the laboratory. *Review of Economic Studies*, 91(5):2806–2831.

Fehr, D., Hakimov, R., and Kübler, D. (2015). The willingness to pay–willingness to accept gap: A failed replication of Plott and Zeiler. *European Economic Review*, 78:120–128.

Frederick, S. (2005). Cognitive reflection and decision making. *Journal of Economic Perspectives*, 19(4):25–42.

Greiner, B. (2015). Subject pool recruitment procedures: Organizing experiments with ORSEE. *Journal of the Economic Science Association*, 1(1):114–125.

Hao, L. and Houser, D. (2012). Belief elicitation in the presence of naïve respondents: An experimental study. *Journal of Risk and Uncertainty*, 44(2):161–180.

Healy, P. J. (2006). Learning dynamics for mechanism design: An experimental comparison of public goods mechanisms. *Journal of Economic Theory*, 129(1):114–149.

Healy, P. J. (2020). Explaining the BDM—or any random binary choice elicitation mechanism—to subjects. *Working paper*.

Kagel, J., Harstad, R., and Levin, D. (1987). Information impact and allocation rules in auctions with affiliated private values: A laboratory study. *Econometrica*, 55(6):1275–1304.

Kagel, J. H. and Levin, D. (1993). Independent private value auctions: Bidder behaviour in first-, second-and third-price auctions with varying numbers of bidders. *The Economic Journal*, 103(419):868–879.

Li, S. (2017). Obviously strategy-proof mechanisms. *American Economic Review*, 107(11):3257–3287.

Mamadehussene, S. and Sguera, F. (2023). On the reliability of the BDM mechanism. *Management Science*, 69(2):1166–1179.

Martin, D. and Muñoz-Rodriguez, E. (2022). Cognitive costs and misperceived incentives: Evidence from the BDM mechanism. *European Economic Review*, 148:104197.

Martínez-Marquina, A., Niederle, M., and Vespa, E. (2019). Failures in contingent reasoning: The role of uncertainty. *American Economic Review*, 109(10):3437–3474.

Miller, G. A. (1956). The magical number seven, plus or minus two: Some limits on our capacity for processing information. *Psychological Review*, 63(2):81.

Möbius, M. M., Niederle, M., Niehaus, P., and Rosenblat, T. S. (2022). Managing self-confidence: Theory and experimental evidence. *Management Science*, 68(11):7793–7817.

Ngangoué, M. K. and Schotter, A. (2023). The common-probability auction puzzle. *American Economic Review*, 113(6):1572–1599.

Niederle, M. and Vespa, E. (2023). Cognitive limitations: Failures of contingent thinking. *Annual Review of Economics*, 15(1):307–328.

Plott, C. R. and Zeiler, K. (2005). The willingness to pay-willingness to accept gap, the "endowment effect," subject misconceptions, and experimental procedures for eliciting valuations. *American Economic Review*, 95(3):530–545.

Sims, C. A. (2003). Implications of rational inattention. *Journal of Monetary Economics*, 50(3):665–690.

Yu, C. W., Zhang, Y. J., and Zuo, S. X. (2021). Multiple switching and data quality in the multiple price list. *Review of Economics and Statistics*, 103(1):136–150.

A Appendix

A.1 Additional Tables and Figures

	(1)	(2)	(3)
Dependent variable: plays optimal strategy, round 2 ($1.75 ≤ bid ≤ $2.25)			
Variables			
BDM (CBC)	0.006	0.008	0.035
	(0.050)	(0.050)	(0.063)
CLOCK (NOT OSP)	0.101	0.100	0.068
	(0.064)	(0.066)	(0.067)
CLOCK (OSP)	0.180***	0.177***	0.161**
	(0.062)	(0.061)	(0.077)
PL (NOT GSO)	0.417***	0.408***	0.414***
	(0.083)	(0.081)	(0.075)
PL (GSO)	0.296***	0.290***	0.321***
	(0.064)	(0.059)	(0.069)
optimal strategy	0.539***	0.542***	0.424***
(round 1)	(0.124)	(0.127)	(0.121)
optimal strategy	0.539***	0.542***	0.424***
(round 1)×			
BDM (CBC)	-0.020	-0.022	-0.025
	(0.174)	(0.176)	(0.159)
CLOCK (NOT OSP)	-0.252	-0.242	-0.133
	(0.215)	(0.207)	(0.195)
CLOCK (OSP)	-0.253	-0.280	-0.271
	(0.371)	(0.379)	(0.361)
PL (NOT GSO)	-0.199	-0.151	-0.109
	(0.157)	(0.164)	(0.151)
PL (GSO)	-0.062	-0.065	-0.058
	(0.176)	(0.162)	(0.148)
maximum bid - 6		-0.048***	-0.041***
		(0.015)	(0.014)
constant	0.167***	0.161***	0.181***
	(0.023)	(0.037)	(0.027)
subject controls	N	N	Y
observations	386	386	386
right-censored	49	49	49
log-likelihood	-195.9	-191.4	-177.4

*** p<0.01, ** p<0.05, * p<0.1

Table A.1: Censored regressions of optimal bid in round 2 on mechanism and whether subject makes an optimal bid in round 1. Right-censored values for the dependent variable are handled using maximum likelihood estimation in Equation (2). Right-censored values for the regressor, optimal strategy (round 1), are predicted from the corresponding model's regression results in Table 4. All regressions utilize cluster-robust standard errors at the session level. Subject controls are relative to mean value and include gender, whether a subject is an econ major, and number of correct answers on CRT.

A.2 Example Decision Card

Alpha

This card is worth $2.00 to you.

You can also sell the card to the computer in front of you at an unknown price.

The unknown price was randomly drawn between $_____ and $_____, in increments of 25 cents. Every possible value has an equal chance of being selected.

Please enter the name on the top right of the card into the computer. You should confirm that the range on the computer matches the range on this card.

The unknown price is pre-determined: nothing you do will change its value. (To make this point clear, the unknown price is also written on the back of this card.)

When the experimenter instructs you to, please press start on the computer. The computer will guide you on how to determine you "bid." Once you make this decision on the computer you can write it down below.

Please name your bid. $_____.

If your bid is below or equal to the unknown price on the back of the card, then you sell your card at your bid.

If your bid is above the unknown price on the back of this card, then you do not sell your card, but you do collect the $2.00 value of the card.

You can only view the unknown price AFTER the computer tells you to do so.

Figure A.1: Front side of the decision card for BDM-based and CLOCK-based mechansisms.

The unknown price is under the tape. To be viewed only after you have named your bid on the other side.

Circle the appropriate amount and print your name so we can pay you.

-My bid is below or equal to the unknown price. Pay me the price of $_____

-My bid is above the unknown price. Pay me $2.00

Subject ID: _____

Figure A.2: Back side of the decision card for BDM-based and CLOCK-based mechansisms.

Alpha

This card is worth $2.00 to you.

You can also sell the card to the computer in front of you at an unknown price.

The unknown price was randomly drawn between $_____ and $_____, in increments of 25 cents. Every possible value has an equal chance of being selected.

Please enter the name on the top right of the card into the computer. You should confirm that the range on the computer matches the range on this card.

The unknown price is pre-determined: nothing you do will change its value. (To make this point clear, the unknown price is also written on the back of this card.)

When the experimenter instructs you to, please press start on the computer. The computer will guide you on how to make decisions.

If you choose to sell the card at the unknown price on the back of this card, then you sell your card at the unknown price.

If you choose to keep the card at the unknown price on the back of this card, then you do not sell your card, but you do collect the $2.00 value of the card.

You can only view the unknown price AFTER the computer tells you to do so.

Figure A.3: Front side of the decision card for PL-based mechansisms.

The unknown price is under the tape. To be viewed only after you have made your decisions on the other side.

Circle the appropriate amount and print your name so we can pay you.

-I sell the card at the unknown price. Pay me the price of $_____

-I keep the card at the unknown price. Pay me $2.00

Subject ID: _____

Figure A.4: Back side of the decision card for PL-based mechansisms.